

Chiral soliton lattice in inhomogeneous magnetic fields

What the QCD anomaly does when the magnetic field bends

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Theory seminar

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The setting: QCD in a field of 10^{19} gauss

Strong- B QCD is not exotic any more:

- magnetar interiors, $B \sim 10^{15}\text{--}10^{19}$ G;
- off-central heavy-ion collisions, the strongest fields ever made;
- lattice QCD in a background field.

The interesting scale is set not by m_ρ^2/e but by the **anomaly**:

$$B_{\text{CSL}} = \frac{16\pi f_\pi^2 m_\pi}{\mu} \approx 0.06 \text{ GeV}^2 \approx 10^{19} \text{ G}$$

which *vanishes in the chiral limit*.

The known story (2007–2017)

Above B_{CSL} , the QCD ground state at finite μ_B is not the vacuum and not nuclear matter: it is a periodic, parity-breaking condensate of *neutral pions* — the **chiral soliton lattice**.

The fine print

Every word of that assumes B is **uniform**.

In every system we care about, it is not.

The story in four steps

I **A wall.** The anomaly gives a π^0 domain wall a baryon number $\propto B$. [Son–Stephanov '07]

II **A lattice.** Solve the EFT exactly: a soliton lattice, with an analytic phase diagram.

[Brauner–Yamamoto '17]

III **A bent field.** What survives when B is inhomogeneous? ← this talk [TB + RR '26]

IV **What we learned**, and what we still cannot do.

The question in one line

Does the CSL need a uniform magnetic field, or only *some* magnetic field?

I. The anomaly builds a wall

Why only π^0 survives

Two-flavour ChPT, $\Sigma \in SU(2)$, in an external magnetic field:

$$\mathcal{L} = \frac{f_\pi^2}{4} \left[\text{Tr}(D_\nu \Sigma^\dagger D^\nu \Sigma) + 2m_\pi^2 \text{Re Tr } \Sigma \right]$$

- Charged pions are reorganised into Landau levels, $E^2 \supset (2n+1)eB$: for $\sqrt{eB} \gg m_\pi$ they are **gapped and decouple**.
- What is left is a single real field, $\Sigma = e^{i\varphi\tau_3}$ — a *sine-Gordon* theory.
- The magnetic field enters that theory only through the **Wess–Zumino–Witten** term (the same one that runs $\pi^0 \rightarrow 2\gamma$).

The reduced EFT — everything below follows from this

$$\mathcal{H} = f_\pi^2 \left[\frac{1}{2} (\nabla\varphi)^2 + m_\pi^2 (1 - \cos\varphi) - \mathbf{H} \cdot \nabla\varphi \right], \quad \mathbf{H} \equiv \frac{\mu\mathbf{B}}{4\pi^2 f_\pi^2}$$

The anomalous term is a total derivative — and that is the point

Since $\nabla \cdot \mathbf{H} = 0$, the term $-\mathbf{H} \cdot \nabla \varphi$

- does **not** enter the equation of motion,
- but **does** enter the energy,
- and it is *linear* in $\nabla \varphi$ — so it always pays to wind φ along $\mathbf{B}$.

The anomalous baryon current gives the winding a physical meaning:

$$n_B = \frac{\mathbf{B} \cdot \nabla \varphi}{4\pi^2}, \quad m = \frac{\mu \nabla \varphi}{4\pi^2}.$$

Son–Stephanov: the π^0 domain wall

$\varphi : 0 \rightarrow 2\pi$ across a wall of thickness $1/m_\pi$:

$$\frac{E}{S} = 8f_\pi^2 m_\pi, \quad \frac{N_B}{S} = \frac{eB}{2\pi}$$

$$\Rightarrow \frac{E}{N_B} = \frac{16\pi f_\pi^2 m_\pi}{eB}$$

Beat nuclear matter ($E/N_B \approx m_N$) when

$$B > B_1 = \frac{16\pi f_\pi^2 m_\pi}{e m_N} \approx 1.1 \times 10^{19} \text{ G}.$$

Locally stable (no tachyonic $\pi^\pm$ on the wall)
above $B_0 = 3m_\pi^2/e \approx 1.0 \times 10^{19} \text{ G}$.

From one wall to a stack

If one wall is cheaper per baryon than nuclear matter, so are two, so are N ...

⇒ the ground state should be a **stack of parallel walls**, their spacing fixed by the competition

$$\underbrace{m_\pi^2(1 - \cos \varphi)}_{\text{likes } \varphi=0} \quad \text{vs.} \quad \underbrace{-\mathbf{H} \cdot \nabla \varphi}_{\text{likes winding}}$$

But a stack is a conjecture

Son–Stephanov did not solve the EoM for the stack.

Two questions were open:

- What *is* the exact ground state?
- Where exactly does it appear in the (μ, B) plane?

That is step II.

II. The lattice, solved exactly

The pendulum in disguise

Static, 1D configurations along $B \parallel \hat{z}$:

$$\partial_z^2 \varphi = m_\pi^2 \sin \varphi \quad \Longleftrightarrow \quad \text{simple pendulum}$$

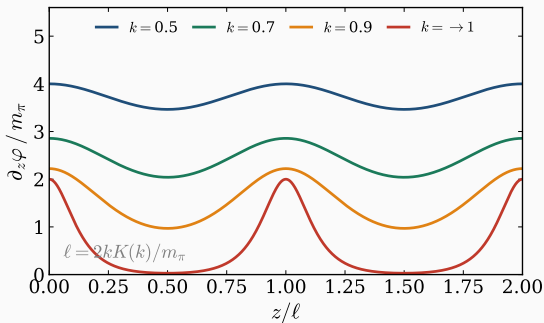
Exact solution in Jacobi elliptic functions,

$$\cos \frac{\varphi(z)}{2} = \text{sn}(\bar{z}, k), \quad \bar{z} = \frac{zm_\pi}{k},$$

period

$$\ell = \frac{2kK(k)}{m_\pi}.$$

- $k \rightarrow 1$: isolated, widely separated walls
- $k \rightarrow 0$: smooth, almost sinusoidal winding



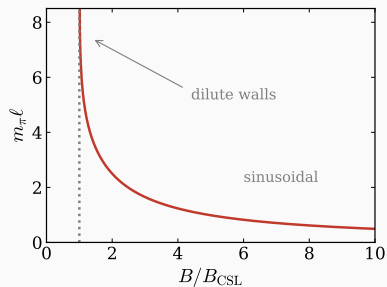
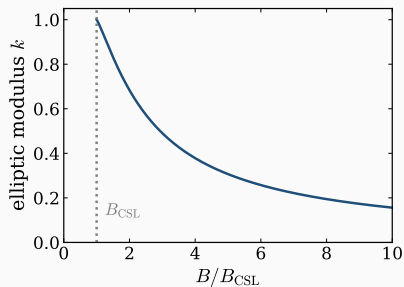
$\partial_z \varphi$ is $\propto$ the baryon density: the lattice is a periodic array of
topological objects.

Selecting the ground state

Average energy density per unit cell:

$$\langle \bar{\mathcal{E}} \rangle = 4 \left[\frac{1}{2} \left(1 - \frac{1}{k^2} \right) + \frac{E(k)}{k^2 K(k)} - \frac{1}{k K(k)} \frac{\bar{H}}{\bar{H}_{\text{CSL}}} \right] \xrightarrow{\partial_k=0} \boxed{\frac{E(k)}{k} = \frac{B}{B_{\text{CSL}}}}, \quad B_{\text{CSL}} = \frac{16\pi f_\pi^2 m_\pi}{\mu}$$

Since $E(k)/k \geq 1$: the CSL exists if and only if $B > B_{\text{CSL}}$.



What makes it a genuine phase

Topological charges per unit cell

$$\frac{N_B}{S} = \frac{B}{2\pi}, \quad \frac{M}{S} = \frac{\mu}{2\pi}$$

depend only on $\varphi(\pm\ell/2)$, not on the profile.

Broken symmetries

- parity,
- continuous translations along B
 $\Rightarrow$ a **phonon** with linear dispersion.

Hierarchical symmetry breaking: chiral symmetry

first, then translations, with

$p_{\text{CSL}} = B/(4\pi^2 f_\pi^2) \ll 4\pi f_\pi$ controlling the expansion.

And an upper edge: fluctuations of $\pi^\pm$ in the lattice obey a Lamé equation ($n=2$),

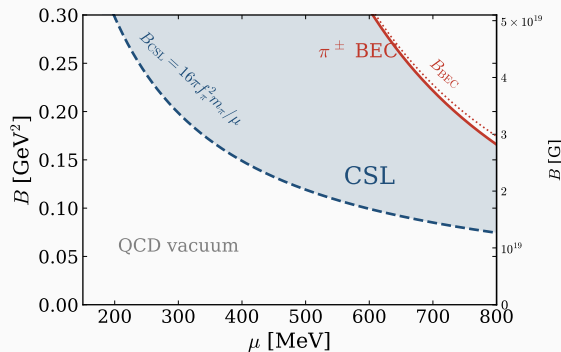
$$\min \omega_{n=0}^2 = B - \frac{m_\pi^2}{k^2} \left(2 - k^2 + 2\sqrt{1 - k^2 + k^4} \right)$$

which turns negative at large B :

$$B_{\text{BEC}} \xrightarrow{m_\pi \rightarrow 0} \frac{16\pi^4 f_\pi^4}{\mu^2}$$

$\Rightarrow$ charged-pion BEC destabilises the CSL from above.

The phase diagram we inherited



- Model-independent: leading-order ChPT + anomaly, accurate to $p_{\text{CSL}}^2 / (4\pi f_\pi)^2$.
- Fully analytic.
- Physical window sits right at magnetar field strengths.

The assumption

Uniform B , infinite volume.

Both are wrong in *every* application: heavy-ion fields are violently inhomogeneous and time-dependent; magnetar fields are dipolar; the lattice is a finite torus.

III. What happens when the field bends

Three reasons to care about non-uniform B

1. **Heavy-ion collisions.** B varies over the transverse size of the fireball and decays on $\sim \text{fm}/c$ timescales.
2. **Compact stars.** Dipolar (or worse) field geometry; the variation is slow on QCD scales, but the *geometry* matters — field lines close.
3. **Lattice QCD.** On a torus the flux is quantised, $eB = 2\pi N/(L_x L_y)$.

The standard workaround [Levkova–DeTar '14]: make B constant on half the lattice and **opposite** on the other half, so the total flux vanishes and B can be tuned continuously.

$\Rightarrow$ the field *used in simulations* is a magnetic domain wall.

Our questions

- Does inhomogeneity destroy the CSL?
- If not: for *which* fields does it survive?
- Can a bent field ever *help*?

Strategy: same EFT, classical minimisation, but now a finite domain V and a general $H(x)$.

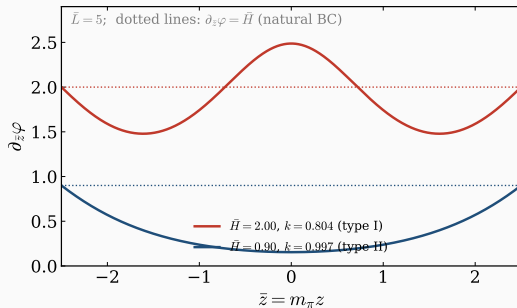
First: a finite box already changes things

Minimise $\bar{E}[\varphi] = \int d\bar{z} [\frac{1}{2}(\partial_{\bar{z}}\varphi)^2 + 1 - \cos\varphi - \bar{H}\partial_{\bar{z}}\varphi]$ on $\bar{z} \in (-\bar{L}/2, \bar{L}/2)$, imposing no boundary condition by hand.

The variational principle supplies its own:

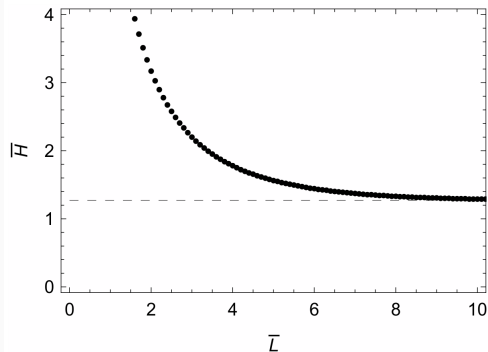
$$\mathbf{n} \cdot \nabla \varphi_0 = \mathbf{n} \cdot \mathbf{H} \text{ on } \partial V$$

- the ground state is **never** uniform for $\bar{H} \neq 0$;
- only a *discrete* set of k is now allowed (vs. a continuum in infinite volume);
- two families: gradient maximal at the centre (I) or minimal (II).



Solutions of the EoM satisfying the natural BC at $\bar{L} = 5$.

The finite-volume phase boundary



Critical field $\bar{H}_{\text{cr}}$ vs. box size $\bar{L} = m_\pi L$. Dashed: infinite-volume value $4/\pi$.

- We define the transition as the flip from family II (bending only at the boundary) to family I (a soliton in the bulk) — the finite-volume analogue of the domain wall that signals the onset.
- $\bar{H}_{\text{cr}}(\bar{L}) \rightarrow 4/\pi \approx 1.27$, i.e. $B \rightarrow B_{\text{CSL}}$, as $\bar{L} \rightarrow \infty$. **Good.**
- A small box *suppresses* the CSL: you need a stronger field to fit a soliton in.

Why this matters

Finite-volume corrections are a prerequisite for comparing with lattice QCD — and for non-uniform fields, where a thermodynamic limit may not even exist.

The chiral limit: the problem becomes linear

Set $m_\pi = 0$. The energy functional $E[\varphi] = f_\pi^2 \int_V [\frac{1}{2}(\nabla\varphi)^2 - \mathbf{H} \cdot \nabla\varphi]$ gives

$$\nabla^2\varphi_0 = 0 \quad \text{in } V, \quad \mathbf{n} \cdot \nabla\varphi_0 = \mathbf{n} \cdot \mathbf{H} \quad \text{on } \partial V.$$

A Neumann problem for the Laplace equation

Solution exists and is **unique up to a constant shift** — so in the chiral limit there is exactly one stationary state, and it is automatically the ground state. No competition, no phase transition.

The whole question “which fields support a CSL?” is now the question:

what does the map $\mathbf{H} \mapsto \varphi_0[\mathbf{H}]$ look like?

The answer: φ_0 is the magnetic scalar potential

Decompose the field on the domain V (gradient-solenoid / Helmholtz):

$$\mathbf{H} = \nabla\psi[\mathbf{H}] + \mathbf{A}[\mathbf{H}], \quad \nabla \cdot \mathbf{A} = 0 \text{ in } V, \quad \mathbf{n} \cdot \mathbf{A} = 0 \text{ on } \partial V.$$

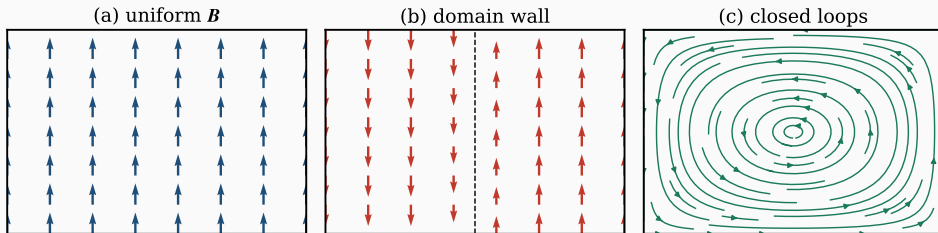
Complete the square in the energy:

$$E[\varphi] = f_\pi^2 \int_V \left[\frac{1}{2}(\nabla\varphi - \nabla\psi)^2 - \frac{1}{2}(\nabla\psi)^2 \underbrace{-\mathbf{A} \cdot \nabla\varphi}_{\text{integrates to 0}} \right]$$

$$\Rightarrow \boxed{E[\varphi] \geq -\frac{f_\pi^2}{2} \int_V (\nabla\psi)^2 \geq -\frac{f_\pi^2}{2} \int_V H^2}$$

- first bound saturated $\iff \nabla\varphi = \nabla\psi$: the ground state is $\psi[\mathbf{H}]$;
- second bound saturated $\iff \mathbf{A} = 0$, i.e. $\mathbf{H}$ is conservative.

Two theorems, and a picture



1. When is there *no* CSL?

$E \leq 0$ always, with equality iff $\nabla\psi = 0$ — exactly when B is **tangential to ∂V everywhere** (c). Any field that pokes through the boundary condenses pions.

2. When is the CSL *optimal*?

The bound $-\frac{f_\pi^2}{2} \int H^2$ is reached iff H is conservative (a). Then $\nabla\varphi_0 = H$ *pointwise*: the gradient **follows the field lines**, whatever the shape of V .

For non-conservative fields (b), both the decomposition and φ_0 depend on the domain — *boundaries are physical*.

A worked family: separation of variables

Take $H(x, z) = (0, H(x))$ on a rectangle $|x| \leq L_x/2$, $|z| \leq L_z/2$:

$$\varphi_0(x, z) = \frac{z}{L_x} \int dt H(t) + \sum_{\text{even } n > 0} \frac{2 \int dt H(t) \cos \frac{\pi n t}{L_x}}{\pi n \cosh \frac{\pi n L_z}{2 L_x}} \cos \frac{\pi n x}{L_x} \sinh \frac{\pi n z}{L_x} + \sum_{\text{odd } n > 0} (\cos \leftrightarrow \sin)$$

For the **magnetic domain wall** $H_{\text{DW}} = H_0 \operatorname{sgn} x$ and as $L_x \rightarrow \infty$ the sum becomes an integral, only the odd sum survives:

$$\varphi_{\text{DW}} = \sum_{\text{odd } n > 0} \frac{4 H_0 L_x}{\pi^2 n^2 \cosh \frac{\pi n L_z}{2 L_x}} \sin \frac{\pi n x}{L_x} \sinh \frac{\pi n z}{L_x} \quad \varphi_{\text{DW}} \rightarrow \frac{H_0}{\pi} \int_{-\infty}^{+\infty} dk \frac{\sin kx \sinh kz}{k^2 \cosh \frac{k L_z}{2}}.$$

These closed forms are what we use to benchmark the numerics later.

Bonus: a non-uniform chemical potential comes for free

If $\mu = \mu(\mathbf{x})$ (local equilibrium in an inhomogeneous system), then $\mathbf{H} = \mu \mathbf{B} / (4\pi^2 f_\pi^2)$ is no longer solenoidal and

$$\nabla^2 \varphi_0 = \nabla \cdot \mathbf{H} = \frac{\mathbf{B} \cdot \nabla \mu}{4\pi^2 f_\pi^2},$$

a *Poisson* equation — with the same natural BC.

- The decomposition $\mathbf{H} = \nabla \psi + \mathbf{A}$ still holds ($\mathbf{A}$ is still divergence-free and tangential), so $\varphi_0 = \psi[\mathbf{H}]$ **still**.
- Neither bound used $\nabla \cdot \mathbf{H} = 0$: every statement survives verbatim, phrased in terms of $\mathbf{H}$ rather than $\mathbf{B}$.
- CSL still beats the vacuum unless $\mathbf{B}$ is tangential to ∂V — *independently of $\mu(\mathbf{x})$* .
- The optimal bound now requires $\mathbf{H}$ conservative: $\nabla \mu \times \mathbf{B} + \mu \nabla \times \mathbf{B} = 0$.

Massive pions: back to non-linearity, but two facts survive

Now $m_\pi \neq 0$ and the EoM is non-linear again. Still, without any computation:

Fact 1 — a no-go

The mass term is non-negative, so $E \geq E|_{m_\pi=0} \geq 0$ for fields tangential to ∂V . **Tangential fields never produce a CSL, at any m_π .**

Fact 2 — a guarantee

Use $\psi[H]$ as a variational trial state:

$$E[\psi] = f_\pi^2 \int_V \left[-\frac{1}{2}(\nabla\psi)^2 + m_\pi^2(1 - \cos\psi) \right]$$

For conservative fields $\nabla\psi = H$, and the integrand is negative everywhere provided $|H| > 2m_\pi$, i.e.

$$|B| > \frac{8\pi^2 f_\pi^2 m_\pi}{\mu} = \frac{\pi}{2} B_{\text{CSL}} \approx 0.09 \text{ GeV}^2$$

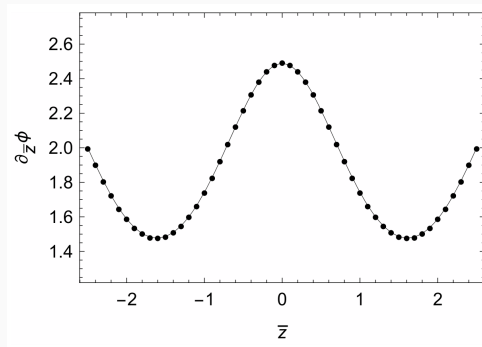
Remarkably close to the uniform-field threshold — and valid for *any* domain V .

Numerics: what we do, and why you should believe it

Method (Mathematica 15, built from scratch)

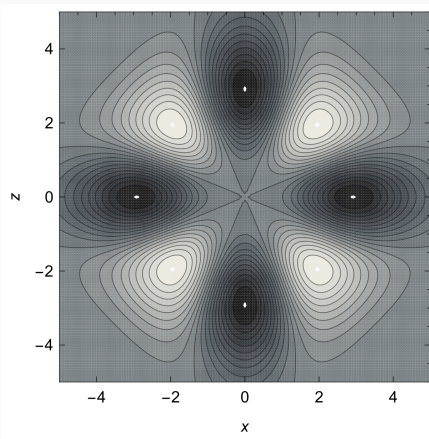
- discretise V on an $n_x \times n_z$ grid;
- central differences in the bulk, one-sided at the edge;
- trapezoidal rule (higher-order integrators develop artifacts on modulated states);
- minimise over *all* grid values — **no boundary condition imposed**, the natural BC emerges;
- NMinimize (global, coarse) cross-checking FindMinimum (local, fine).

Validation: scan $0 < \bar{L} \leq 10$, $0 < \bar{H} \leq 5$ on a 50×50 grid: the code finds the true global minimum at **2495 of 2500** points (the five misses sit on curves of degenerate minima).



1D benchmark, $\bar{L} = 5$, $\bar{H} = 2$: points = free minimisation on 50 cells, line = analytic solution. Note $\partial_{\bar{z}}\varphi = \bar{H} = 2$ at the edges — the natural BC, found, not imposed.

Two dimensions: a Gaussian blob of field

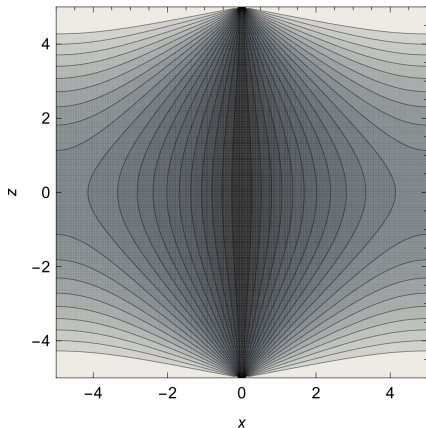


Field from a Gaussian scalar potential, $\psi = \psi_0 e^{-(x^2+z^2)/R^2}$, $H = (-\partial_z \psi, \partial_x \psi)$, in the chiral limit on a 10×10 box.

- Plotted: the anomalous baryon density $H \cdot \nabla \varphi$.
- Bright lobes = positive baryon density, dark = negative.
- The condensate *textures itself* to the field: the lattice is no longer a stack of flat planes, it is a curved, locally-defined object.

This is an honest 2D minimisation on a 51×51 grid — no ansatz.

The case study: a magnetic domain wall



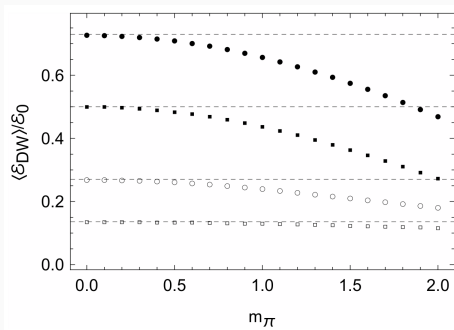
$H_\xi(x) = H_0 \tanh(x/\xi)$: constant field of opposite sign on the two halves — exactly the lattice-QCD trick.

- The CSL **survives**, even though this field is *not* conservative.
- Baryon density vanishes on the wall at $x = 0$ and is symmetric about it.
- ξ interpolates between a sharp wall ($\xi \rightarrow 0$) and a field effectively linear in x ($\xi \gtrsim L_x$).

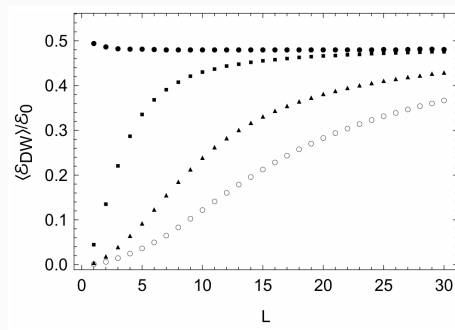
Benchmark

Chiral limit, square domain, sharp wall: analytics predict $\langle \mathcal{E}_{\text{DW}} \rangle = \mathcal{E}_0/2$ exactly; the code gives $0.499 \mathcal{E}_0$ on a 31×31 grid, independently of L .

How much does bending cost?



Energy vs. pion mass, for aspect ratios
 $\alpha = L_z/L_x = 0.5, 1, 2, 4$ (sharp wall, $H_0 = 5$).



Energy vs. system size, for wall widths $\xi \rightarrow 0, 1, 3, 5$
($m_\pi = 0.5$).

- $m_\pi \rightarrow 0$ intercepts reproduce the analytic values (dashed) — the code is right.
- Increasing m_π monotonically *reduces* the condensation energy: the mass term fights the winding.
- Condensation switches off for $L \lesssim \xi$ and saturates for $L \gg \xi$.

The analytic gem: how much energy lives *on* the wall?

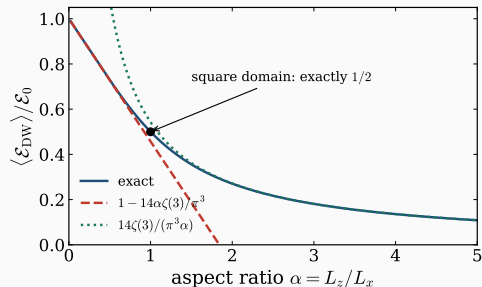
In the chiral limit the domain-wall energy can be summed in closed form:

$$\frac{\langle \mathcal{E}_{\text{DW}} \rangle}{\mathcal{E}_0} = \frac{16}{\alpha} \sum_{\text{odd } n > 0} \frac{1}{(\pi n)^3} \tanh \frac{\pi n \alpha}{2}, \quad \alpha = \frac{L_z}{L_x}$$

Using the **fermionic Matsubara identity** $\sum_{\text{odd } k > 0} (k^2 + t^2)^{-1} = \frac{\pi}{4t} \tanh \frac{\pi t}{2}$:

$$\frac{\langle \mathcal{E}_{\text{DW}} \rangle}{\mathcal{E}_0} = 1 - \frac{14\alpha\zeta(3)}{\pi^3} + \frac{32\alpha}{\pi^3} \sum_{\text{odd } k > 0} \frac{1}{k^3} \frac{1}{e^{\pi k/\alpha} + 1}$$

- the first two terms are the *entire* power series in α ;
- everything else is exponentially small;
- $\alpha \rightarrow 0$: the uniform-field answer, $\mathcal{E}_0$;
- $\alpha = 1$: exactly $\mathcal{E}_0/2$.



Linear in $\alpha \Rightarrow$ energy excess $\propto L_x^0 L_z^2$: **localised on the wall**.

IV. Where this leaves us

1. Inhomogeneity does not kill the CSL — geometry decides

B tangential to the boundary everywhere $\Rightarrow$ no condensation, at any pion mass. Anything else $\Rightarrow$ the vacuum is unstable.

2. Conservative fields are the friendly case

Chiral limit: $\nabla\varphi_0 = H$ pointwise, the energy saturates its bound and does not care about the domain. Massive: a CSL is *guaranteed* for $|B| > \frac{\pi}{2} B_{\text{CSL}} \approx 0.09 \text{ GeV}^2$, for any V .

3. Bending costs, but not much, and it is a surface effect

For the lattice-QCD domain-wall field, the cost is exactly computable in the chiral limit and localised on the wall: $\langle \mathcal{E}_{\text{DW}} \rangle / \mathcal{E}_0 = 1 - 14\alpha\zeta(3)/\pi^3 + \dots$

Along the way: the finite-volume CSL phase boundary, and a working 2D energy minimiser with no boundary conditions imposed by hand.

What we cannot do yet

Excitations. The ground state is one property; transport and thermodynamics need the spectrum. In a uniform field it is a Lamé problem. In a bent field, the phonon is no longer a plane wave — and it is not even obvious what the Nambu–Goldstone counting is.

Realistic fields. We used mathematically clean profiles. A dipole field, or an event-by-event heavy-ion field, is the honest next step — and would turn this into a phenomenological statement.

Beyond the classical minimisation.

Thermal and quantum fluctuations shift the phase boundary already in the uniform case [Brauner–Kolešová–Yamamoto '21; NLO '23]. With inhomogeneity, that computation is open.

And a different attack. The sine-Gordon-plus-anomaly Hamiltonian discretises cleanly onto a qubit register (clock/shift operators, Π^2 from the QFT). The magnetic term needs ≥ 2 qubits per site to be non-trivial — first steps towards simulating the real-time formation of a CSL.
(*backup*)

Thank you

Bending the magnetic field costs condensation energy,
but only closing the field lines destroys the lattice.

Backup: finite-volume energies

With $\theta = \varphi - \pi$, $\theta = 2 \operatorname{am}(\frac{\bar{z}-\bar{z}_0}{k}, k)$, the energy on $\bar{z} \in (-\bar{L}/2, \bar{L}/2)$ is

$$\bar{E}(k, \bar{z}_0) = 2\bar{L}\left(1 - \frac{1}{k^2}\right) + \frac{4}{k}\left[\mathcal{E}\left(\frac{\bar{z}-\bar{z}_0}{k}, k\right)\right]_{-\bar{L}/2}^{+\bar{L}/2} - 2\bar{H}\left[\operatorname{am}\left(\frac{\bar{z}-\bar{z}_0}{k}, k\right)\right]_{-\bar{L}/2}^{+\bar{L}/2}$$

Family I ($\bar{z}_0 = 0$, gradient maximal at the centre):

$$\bar{E}_{\max}(k) = 2\bar{L}\left(1 - \frac{1}{k^2}\right) + \frac{8}{k}\mathcal{E}\left(\frac{\bar{L}}{2k}, k\right) - 4\bar{H}\operatorname{am}\left(\frac{\bar{L}}{2k}, k\right)$$

Family II ($\bar{z}_0 = kK(k)$, gradient minimal at the centre):

$$\bar{E}_{\min}(k) = 2\bar{L}\left(1 - \frac{1}{k^2}\right) + \frac{8}{k}\mathcal{E}\left(\frac{\bar{L}}{2k}, k\right) - 8k \operatorname{sn} \operatorname{cd}\left(\frac{\bar{L}}{2k}, k\right) - 2\bar{H}\left[\operatorname{am}\left(\frac{\bar{L}}{2k} + K, k\right) + \operatorname{am}\left(\frac{\bar{L}}{2k} - K, k\right)\right]$$

$\mathcal{E}$ = Jacobi epsilon function. The natural BC selects a discrete set of k ; the ground state is the one of lowest $\bar{E}$.

Backup: charged pions in the lattice

Fluctuation equation for π^+ in the CSL background:

$$\left[-\partial_z^2 + 6k^2 \text{sn}^2(\bar{z}, k) \right] \tilde{\pi}_+ = \frac{k^2}{m_\pi^2} \left[\omega^2 - (2n+1)B \right] \tilde{\pi}_+ + (k^2 + 4) \tilde{\pi}_+$$

a Lamé operator with $n = 2 \Rightarrow$ three bands; the lowest band edge gives

$$\min \omega_{n=0}^2 = B - \frac{m_\pi^2}{k^2} (2 - k^2 + 2\sqrt{1 - k^2 + k^4}).$$

- At $k \rightarrow 1$ (single wall) this is $B - 3m_\pi^2$: Son–Stephanov's B_0 .
- At large B , $k \propto 1/B$, so the second term $\sim B^2$: BEC always wins eventually.
- Chiral limit: $B_{\text{BEC}} = 16\pi^4 f_\pi^4 / \mu^2$.

Backup: discretisation control

The EFT has one intrinsic scale, the pion Compton wavelength $1/m_\pi$. A cell of size $L_x/(n_x - 1) \times L_z/(n_z - 1)$ is safe when

$$n_x \gtrsim m_\pi L_x, \quad n_z \gtrsim m_\pi L_z.$$

Example. Uniform field, $H_0 = 5$, $m_\pi = 0.5$, $L = 20$, 31×31 grid:

$$\langle \mathcal{E} \rangle_{\text{num}} = -12.36 \quad \text{vs.} \quad \langle \mathcal{E} \rangle_{\text{exact}} = -12.25 \quad (\sim 1\%).$$

The sharp-domain-wall benchmark in the chiral limit gives $0.499 \mathcal{E}_0$ against the exact $0.5 \mathcal{E}_0$ (0.2% on a 31×31 grid), improving with refinement.

Caveat we state in the paper

Do not vary m_π , L , ξ at fixed grid size without rechecking these inequalities — the continuum limit is expensive because the number of optimisation variables grows as $n_x n_z$.

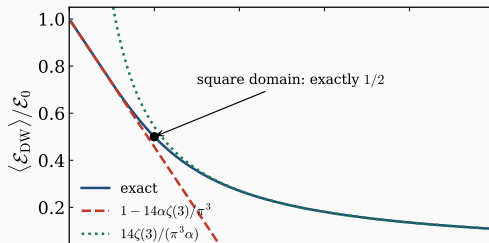
Backup: the wide-domain limit

From the exact representation,

$$\frac{\langle \mathcal{E}_{\text{DW}} \rangle}{\mathcal{E}_0} \xrightarrow{\alpha \rightarrow \infty} \frac{14\zeta(3)}{\pi^3 \alpha},$$

so the total energy scales as $L_x^2 L_z^0$ — in the opposite (thin-slab) limit $\alpha \rightarrow 0$ the deficit scales as $L_x^0 L_z^2$.

Physically: for a tall thin box the wall dominates the whole sample; for a wide flat box the wall is a negligible perturbation on a uniform-field CSL.



Backup: towards quantum simulation of the CSL

Discretise the same Hamiltonian on N sites, $Q = 2^{n_q}$ field values per site, $\hat{Z}|k\rangle = \omega^k|k\rangle$, $\hat{X}|k\rangle = |k+1\rangle$:

$$\hat{H} = t \sum_n \hat{\Pi}_n^2 + J \sum_{\langle nm \rangle} (1 - \cos \Delta\varphi) + h \sum_n (1 - \cos \varphi_n) - \kappa \sum_{\langle nm \rangle} \sin \Delta\varphi$$

with $t = \frac{1}{2f_\pi^2 a}$, $J = \frac{f_\pi^2}{a}$, $h = f_\pi^2 m_\pi^2 a$, $\kappa = f_\pi^2 H = \frac{\mu B}{4\pi^2}$ (a -independent — it is topological).

Exact for any Q : $\cos \varphi \rightarrow \frac{1}{2}(\hat{Z} + \hat{Z}^\dagger)$,
 $\cos \Delta\varphi \rightarrow \frac{1}{2}(\hat{Z}_{n+1}\hat{Z}_n^\dagger + \text{h.c.})$, and

$$\hat{\Pi}^2 = \frac{Q^2 + 2}{12} 1 + \sum_{m=1}^{Q-1} \frac{(-1)^m}{2 \sin^2(\pi m/Q)} \hat{X}^m$$

A physics constraint on the encoding

For $n_q = 1$, $\Delta\varphi \in \{0, \pm\pi\}$ so $\sin \Delta\varphi \equiv 0$:
the magnetic term *drops out identically*.

The CSL first exists at 2 qubits per site.

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